Ch:1 The foundation (logic and proof)

Propositional → is a statement that's → answer is either (True / false)

→ Declarative statement that is either T / F but not both.

1) Lahore is capital of punjab T
2) $2+3 = 5$ T
3) $2+3 = 6$ F

} Propositions

4) How are you?
5) Read it carefully?
6) $x+3 = 7$?
↳ dependent on x.

} not proposition

Propositional variables are denoted by (p, q, r, s)

negation of p donated by not p, $\neg p$, $\tilde{p}$

P: Today is raining

$\neg p$: Today is not raining.

p	$\neg p$
T	F
F	T

$P \wedge q$ → compound proposition. (And) → a.k.a conjuction.

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

$P \vee q$ → disjunction (OR)

P	q	$P \vee q$
T	T	T
F	T	T
T	F	T
F	F	F

$P \oplus q$ → XOR Same → 0 | false
diff → 1 | true

P	q	$P \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

P → A student can have Salad with dinner

q → A student can have soup with dinner.

$P \oplus q$ → A student can have Salad or Soup with dinner BUT not Both.

OR (Inclusive OR)

→ Both of Statements can be T.

XOR (exclusive OR)

→ only one of the Statements can be true but not Both.

eg I will use all my savings to travel to europe or to buy an electric car

in propositional logic

$P \oplus q$ ← only one of them can be True because all his savings will be used eventually

$P \vee q$ → I will use all my savings to travel to europe or to buy an electric car or both

→ combining of one or more proposition with the help of an operator is called compound proposition.

note: Sometimes in place of and, But is used. to define $P \wedge q$

Q find the disjunction of proposition p and q.

p → Rebecca's PC has more than 16 GB free hard disk space

q → the processor in Rebecca's PC runs faster than 1 GHz.

$p \vee q$ → Rebecca's PC has more than 16GB free hard disk space or the processor in Rebecca's PC runs faster than 1GHz.

Conditional Statements: $p \rightarrow q$ (Implication)

The conditional statement that p implies q $(p \rightarrow q)$ is the proposition "if p then q."

$p \rightarrow q$ is a conditional statement that connects two propositions.

Lets define

p = "3 is an odd number" → T

q = "2 is greater than 5" → F

$p \rightarrow q$ if 3 is an odd no then 2 is greater than 5 → (F)

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

(P does not imply q)

promise not activated hence by default T.

---

p: Ali learns good programming skills
q: Ali will find a good job

p → q if Ali learns good programming skills then Ali will find a good job.

p → q if Ali does learn good programming skills then Ali will not find a good job.

p → q means if p is true then Q must be true aka if () then ().
(Always)

↳ one-way implication

(But if Q is true that does not mean P is true.

cannot go backwards.)

→ P: A no is divisible by 4

Q: The no. is even

P → Q : if a no. is divisible by 4 then the no is even

Q → P : if no is even, then it is divisible by 4.

↳ false ($6/4 \neq 0$)

6 is a even no, and it is not divisible by 4.

note in discrete mathematics.

:= assignment

≡ logically equalent

---

necessary (min req) if p then q

q must be true for p to be true

q is necessary for p.

If I become rich then I will be happy

my happiness is necessary consequence of me being rich

Sufficent :

if p is true, then q is guranteed.

So p is enough to ensure q happens.

p is Sufficent for q to be guranteed

if it rains then the ground is wet

wet ground is necessary if it rained

Rain is Sufficent to make the ground wet

$p \to q$: p is sufficient for q and q is necessary for p

$p \leftrightarrow q$: p and q are both necessary and sufficient for each other.

$\downarrow$

$(p \to q) \wedge (q \to p)$

- p = the **condition** (the "if" part).
- q = the **result** (the "then" part).
- $p \to q$ = the **whole promise**: *"If p, then q."*

Example:

- p: I bring cake.
- q: I will share it.
- $p \to q$: *"If I bring cake, then I will share it."* → **That is the promise.**

Now check:

- If I bring cake ($p = T$) but don't share ($q = F$) → ❌ promise broken.
- If I bring cake and share → ✅ promise kept.
- If I don't bring cake ($p = F$) → promise never activates → not broken → treated as true.

👉 So the **promise is the implication ($p \to q$) itself**, not just p.

Do you want me to draw you a tiny **diagram flow (like if-condition in code)** to show this visually?

+ Ask anything

Forming three propositions from $p \to q$

1) $q \to p$ (converse) [opposite one-way]

2) $\neg q \to \neg p$ (contrapositive)

3) $\neg p \to \neg q$ (inverse)

$p \to q$

note:: Every Implication is logically equivalent to its contrapositive

note:

$p \wedge q$

$q \wedge p$ are the same

Other ways to represent bi-cond

→ if p then q and conversely,

Bi-conditional is both necessary and sufficient

aka bi implication

Bi condition :: $p \leftrightarrow q$ → p if and only if q

True when both p and q have the same values

p	q	R
T	T	T
F	F	T

eg

You will pass the course if and only if you score at least 50%.

if you pass → you must have 50%.

if you have 50% → you will pass

Both p and q happen together

You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old

p : cannot ride the roller coaster
q : under 4 feet tall
r : older than 16 years old

How to identify propositions: Look for the if-then relationship.

cause (p) and effect ($q \vee$)

Look for the words if, whenever, when, only if, necessary, sufficient....

Propositional equivalences

These three are compound propositions.

Tautology: A statement that is always true no matter what.

contradiction: A statement that is always false no matter what.

contingency: A statement that is either True | false depends.

T·T

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logically equivalent → Both propositions have same T·T final values.

notation $p \equiv q$

De-morgan Law

$\neg(p \wedge q) \equiv \neg p \vee \neg q$

$\neg(p \vee q) \equiv \neg p \wedge \neg q$

Laws

Identity Law

1) $p \wedge T \equiv p$

$p \vee F \equiv p$

$\neg(\neg p) \equiv p$

T: True

F: false

P: Any proposition

$\wedge$: And

$\vee$: OR

2) Domination Laws

$p \vee T \equiv T$

$p \wedge F \equiv F$

3) commutative Law

$p \vee q \equiv q \vee p$ (order doesn't matter)

$p \wedge q \equiv q \wedge p$

4) Idempotent Laws → apply when both inputs are same.

$p \vee p \equiv p$

$p \wedge p \equiv p$

5) Associative Laws $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

if we are And-ing three things order doesn't matter.

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

if we are OR-ing three things order doesn't matter.

6) Distributive Laws

$a \times (b + c) = a \times b + a \times c$

$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ → And distributes over OR

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ → OR distributes over AND.

7) Absorption Law

$$p \vee (p \wedge q) \equiv p$$
$$p \wedge (p \vee q) \equiv p$$

8) negation Law

$p \wedge \neg p \rightarrow F$

$p \vee \neg p \rightarrow T$

Logical equivalences involving conditional Statements:

Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent.

p	q	$p \to q$	$\neg(p \to q)$	$\neg q$	$p \wedge \neg q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

Logically equivalent

Q Show that $[p \wedge (p \to q)] \to q$ this compound proposition is a tautology

$[p \wedge (p \to q)] \to q$

p $\quad$ q

logical equivalences are written in Book

$p \to q \equiv \neg p \vee q$

$\neg[p \wedge (p \to q)] \vee q$

apply de-morgan

$\neg p \vee \neg(p \to q) \vee q$

Apply $p \to q \equiv \neg p \vee q$ again.

$\neg p \vee \neg(\neg p \vee q) \vee q$

De. morgan's law

$\neg p \vee (p \wedge \neg q) \vee q$ Distributive Law

$A = \neg p \quad B = p \quad C = \neg q$

$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$

$[(\neg p \vee p) \wedge (\neg p \vee \neg q)] \vee q$

→ Identity Law hence we dropped T.

$[(T) \wedge (\neg p \vee \neg q)] \vee q$

$\neg p \vee \neg q \vee q$ → Associative / $\neg q \vee q = T$

$\neg p \vee T$ (Domination Law) = (T)

for, whereas is equalient to if

But, nevertheless is equalient to AND

Satisfability:: finding ONE T/F value for the variables that make the entire proposition True.

Unsatisfability:: there is no possible way to assign T/F values that make the entire proposition to be True. It is always false.

if a proposition is Always false → taking ¬ will lead to tatology

↑ highest order of precedence

{ }

¬

∧

∨

⋮

use de-morgan law to express the negations of

"Miguel has a cellphone and he has a laptop computer" and

"Heather will go to the concert or Steve will go to concert".

P:. Miguel has a cellphone

q:. he has a laptop computer

r:. Heather will go to the concert

S:. Steve will go to concert

$p \wedge q$ → first idea.

$\neg(p \wedge q)$ → equivalent $\equiv \neg p \vee \neg q$

$\neg(p \vee q) \equiv \neg p \wedge \neg q$

$\neg(p \wedge q) \equiv \neg p \vee \neg q$

$r \vee S$ → Second idea

$\neg(r \vee S)$ → equivalent $\equiv \neg r \wedge \neg S$

"Miguel does not have a cellphone or he does not have a laptop computer"

"Heather will not go to the concert and Steve will not go to concert".

Predicates: is a sentence with Blanck that becomes true | false when you fill in the blank.

Written as $P(x)$ x is a variable.

"x is greater than 5" $x = 7$ $P(7) =$ true → proposition.

$x = 5$ $P(5) =$ false

A predicate is a statement about something that can be T|F depending on the value of the variable. (once we plug in a value it becomes a proposition (T|F)

Quantification: → It is used with predicates to tell how many elements apply to it. tells whether the predicate is true for all, some, none elements

---

universal quantifier (∀)

meaning for all, every

$\forall n\, P(n)$ is true if $P(x)$ is true for all values of x

$\forall n P(n)$

is False when there is an n for which $P(x)$ is false

$Q(n)$: $n+1 > x$

$n = 0$ $\quad 1 > 0$

$n = -1$ $\quad 0 > -1$

$n = \frac{1}{2}$ $\quad \frac{3}{2} > \frac{1}{2}$

$\forall n\, P(n)$: True

$P(n)$: $n > 2$

$\forall n\, P(n)$: false

$P(n)$ → Predicate

Existential Quantification: $\exists$

meaning there exist, at least one

$\exists x\, P(x)$ Read as: There exist atleast one x for which $P(x)$ is true

$P(n)$: $n > 3$

$\exists n\, P(n)$, True

$Q(n)$: $n = n+1$ $\quad n = 2$ $\quad 2 = 2+1$

$\exists n\, P(n)$ = false

Suppose: if all values are true the still $\exists_n P(n)$ is true

Unique Quantification: $\exists!$ there is one and only one value of x that makes $P(x)$ true.

(Exactly one)

$P(n)$: $n-1=0$

$n=1$

$\exists!_n P(n)$ = True

---

logical equivalence:

$\forall_n (P(x) \wedge Q(x)) \equiv (\forall_n P(x)) \wedge (\forall_n Q(x))$

↳ Distribution of $\forall$ over $\wedge$

LHS

For every x both $P(x)$ and $Q(x)$ is True.

RHS

For every x $P(x)$ is True and for every x $Q(x)$ is True

Random examples: Suppose domain of $P(x,y)$ for x is 1,2,3 and y is 1,2,3

→ write out these propositions using disjuctions / conjuctions.

$P(x,y)$ → truth depends on x and y values

(a) $\exists x P(x,3) \rightarrow P(1,3) \vee P(2,3) \vee P(3,3)$

(b) $\forall x \neg P(x,2) \rightarrow \neg P(1,3) \wedge \neg P(2,3) \wedge \neg P(3,3)$

↳ for every possible hence we use ∧

$\forall x P(x,2)$ is not true

Eg $Q(x)$ is $x+1 > 2x$

$1 > 2x - x$

$1 > x$

$x < 1$

Solve $\exists x \neg Q(x)$ and $\forall x \neg Q(x)$

↓

there exist some value(s) for which x is for which $Q(x)$ is not true / $\neg Q(x)$ is true

$\neg(x < 1)$ → negation

$x \geq 1$ → $\neg Q(x)$

$1 \geq 1$ True

$\forall x \neg Q(x)$ for every value of x $\neg Q(x)$ is true

$x \geq 1$

$0 \geq 1$ False

Ex 1.3

60, 35, 31

$\forall x P(x) \vee A$

↓

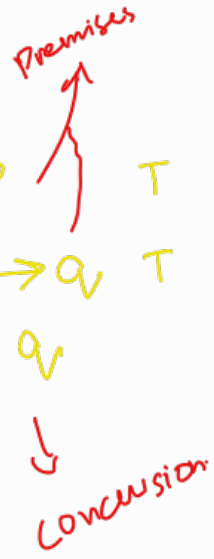
A is a proposition here T/F.

$\exists x \neg Q(x)$

$x \geq 1$

$1 \geq 1$ True

Rules of Inference:. is a form of argument that is valid.
Premises are the given statements from which we derive the conclusion.



1) p T
$p \to q$ T
$\therefore q$

| Tautology: $p \wedge [p \to q] \to q$ | Name: Modus Ponens |

if you have current password then you can login in the system $(p \to q)$

you have a current password. (p)

p:. you have a current password

q:. you can login in the System

$p \to q$
p
$\therefore q$ you can login the System ✓

2) $p \to q$
$\neg q$
$\therefore \neg p$

| $[\neg q \wedge (p \to q)] \to \neg p$ | Modus Tollens |

3) $p \to q$
$q \to r$
$\therefore p \to r$

| $[[p \to q] \wedge [q \to r]] \to [p \to r]$ | H.S |

Syllogism word comes when linking two premises together.

4) $p \vee q$, $\neg p$, $\therefore q$ | $[(p \vee q) \wedge \neg p] \rightarrow q$ | Disjunctive Syllogism.

↳ means if $p \vee q$ is T and $\neg p$ is T then q must be T.

note: For rule of inference to work all the premises must be T.

5) $p \rightarrow T$, $\therefore p \vee q \rightarrow T$ | $p \rightarrow (p \vee q)$ | Addition

no matter what q.

6) $p \wedge q \rightarrow$ Both T, $\therefore p$ OR we can write $\therefore q$ | $(p \wedge q) \rightarrow p$ | Simplification

7) p T, q T, $\therefore p \wedge q$ T | $(p) \wedge (q) \rightarrow p \wedge q$ | conjunction.

8) $p \vee q$

$\neg p \vee r$

$\therefore q \vee r$

$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow q \vee r$ — Resolution.

$p \rightarrow T$

$p \vee q \rightarrow T$ (p is T)

$\neg p \vee r \rightarrow T$ (r is T)

$q \vee r \rightarrow T$ (r is T)

$p \rightarrow F$

$p \vee q \rightarrow T$ (q is T)

$\neg p \vee r \rightarrow T$ (p is T)

$q \vee r \rightarrow T$ (q is T)

Fallacy → doesn't follow Rule of Inference

if it rains (p) the ground gets wet (q).

The ground is wet

$\therefore$ It rains

$p \rightarrow q$

q (✓)

$\therefore p$

$p \rightarrow q$
$p \quad \therefore q$

Rules of Inference for Quantified Statements

Universal Instantiation $\forall x\, P(x) \quad \therefore P(c) \rightarrow T$

if Something is true for all elements then it is true for any specific element

P(c) for any arbitary c

$\therefore \forall x\, P(x)$ → T

if you can prove that a statement is true for a arbitary element then the statement is true for all elements

universal generalisation

arbitary → any no.

$\exists x\, P(x)$

$\therefore P(c)$ for some c → T

if atleast one element satisfy the property.

existential instantiation

P(c) for some c

$\therefore \exists x P(x)$ → T

existence generalisation.

e.g Everyone in discrete mathematics class has taken a course in computer science and Marla is a student in this class imply the conclusion Marla has taken a course in computer Science.

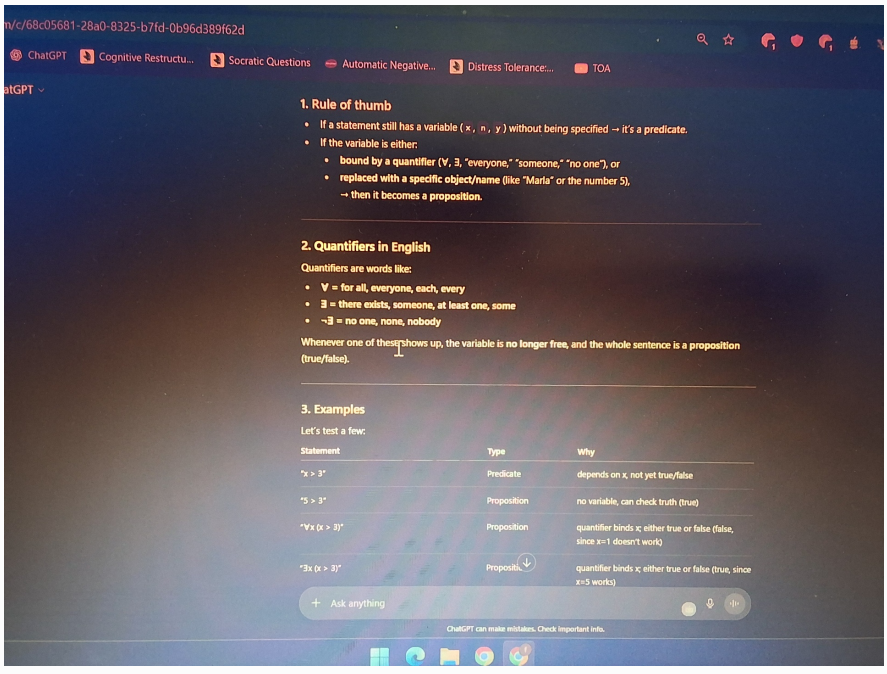
D(x) Everyone in discrete mathematics class

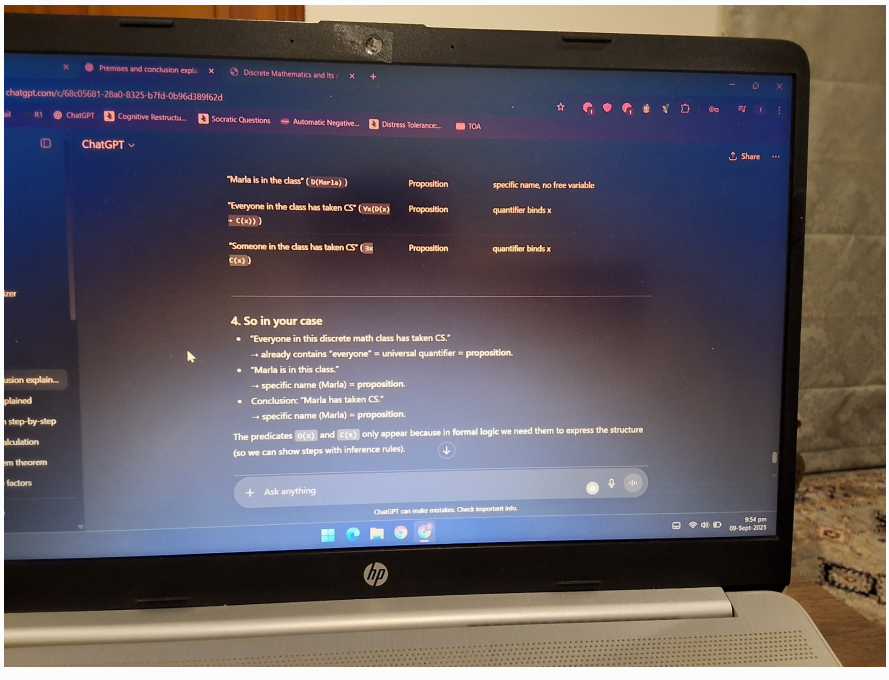
C(x) taken a course in computer science

$\forall x\,(D(x) \rightarrow C(x))$

$\forall x\,(D(\text{Marla}) \rightarrow C(\text{Marla}))$

Predicates are e.g D(x) | C(x) → free variable for x in.





we apply predicates whenever we want to express a property that can apply to many diff things.

Not just one fixed case

D(x) → x is a variable